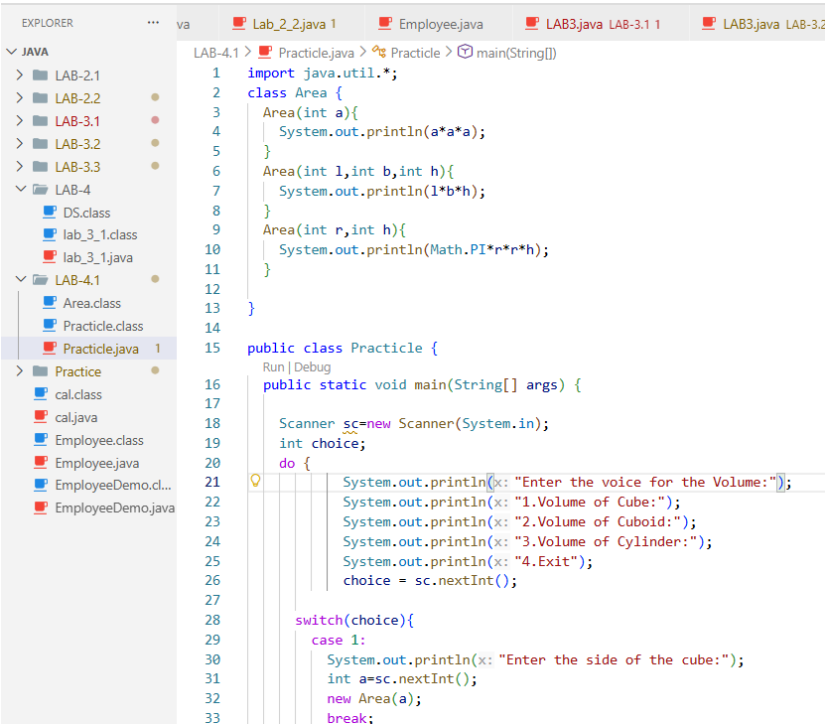
Practical:3.1

Aim:

1.Calculate volume of Cube, Cuboid and Cylinder using Constructor Overloading. (Hint: Constructor Function Name Same but number of arguments will be different. Volume of Cube = a*a*a, Volume of Cuboid = L*B*H, Volume of Cylinder = PI*r*r*h) and Display below Menu, create below constructor functions and call it based on user's choice, get values from user:

1. Calculate Volume of Cube
2. Calculate Volume of Cuboid
3. Calculate Volume of Cylinder
4. Copy Object - Copy Constructor
5. Exit

Code:



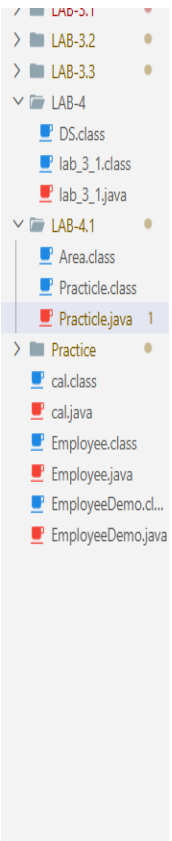

```
import java.util.*;
class Area {
  Area(int a){
    System.out.println(a*a*a);
  }
  Area(int l,int b,int h){
    System.out.println(l*b*h);
  }
  Area(int r,int h){
    System.out.println(Math.PI*r*r*h);
  }

}

public class Practicle {
  Run | Debug
  public static void main(String[] args) {

    Scanner sc=new Scanner(System.in);
    int choice;
    do {
        System.out.println(x: "Enter the voice for the Volume:");
        System.out.println(x: "1.Volume of Cube:");
        System.out.println(x: "2.Volume of Cuboid:");
        System.out.println(x: "3.Volume of Cylinder:");
        System.out.println(x: "4.Exit");
        choice = sc.nextInt();

      switch(choice){
        case 1:
          System.out.println(x: "Enter the side of the cube:");
          int a=sc.nextInt();
          new Area(a);
          break;
```




```
        switch(choice){
          case 1:
            System.out.println(x: "Enter the side of the cube:");
            int a=sc.nextInt();
            new Area(a);
            break;
          case 2:
            System.out.println(x: "Enter the length, breadth and height of the cuboid:");
            int l=sc.nextInt();
            int b=sc.nextInt();
            int h=sc.nextInt();
            new Area(l,b,h);
            break;
          case 3:
            System.out.println(x: "Enter the radius and height of the cylinder:");
            int r=sc.nextInt();
            int h1=sc.nextInt();
            new Area(r,h1);
            break;
            case 4:
            System.out.println(x: "Exiting the program.");
            System.exit(status: 0);
          default:
            System.out.println(x: "Invalid choice! Please enter a valid option.");
        }
      } while (choice!=4);
    }

  }
```

OUTPUT:

```
        at lab_3_1.main(lab_3_1.java:99)
PS C:\Users\AKSHITA\Desktop\JAVA\LAB-4> cd ..
PS C:\Users\AKSHITA\Desktop\JAVA> cd Lab-4.1
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> javac Practicle.java
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cubevoid:
3.Volume of Cylinder:
1
Enter the side of the cube:
5
125
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cubevoid:
3.Volume of Cylinder:
1
Enter the side of the cube:
5
125
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cubevoid:
3.Volume of Cylinder:
1
Enter the side of the cube:
5
125
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cubevoid:
3.Volume of Cylinder:
1
Enter the side of the cube:
```

```
5
125
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cubevoid:
3.Volume of Cylinder:
1
Enter the side of the cube:
5
125
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cubevoid:
3.Volume of Cylinder:
1
Enter the side of the cube:
5
125
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> javac Practicle.java
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1> java Practicle
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cuboid:
3.Volume of Cylinder:
4.Exit
1
Enter the side of the cube:
5
125
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cuboid:
3.Volume of Cylinder:
4.Exit
2
Enter the length, breadth and height of the cuboid:
5 2 8
80
```

```
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cuboid:
3.Volume of Cylinder:
4.Exit
3
Enter the radius and height of the cylinder:
5 3
235.61944901923448
Enter the voice for the Volume:
1.Volume of Cube:
2.Volume of Cuboid:
2.Volume of Cuboid:
3.Volume of Cylinder:
4.Exit
4
3.Volume of Cylinder:
4.Exit
4
Exiting the program.
PS C:\Users\AKSHITA\Desktop\JAVA\Lab-4.1>
```